# MSG No. 172.125 $P6_4$ [ Type I, hexagonal ]

Table 1: Wyckoff site: `3a`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4]` |
| 2 | $\left[0,\ 0,\ z+\frac{2}{3}\right]$ | `[2,5]` |
| 3 | $\left[0,\ 0,\ z+\frac{1}{3}\right]$ | `[3,6]` |

Table 2: Wyckoff site: `3b`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[1,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{3}\right]$ | `[3,6]` |

Table 3: Wyckoff site: `6c`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x-y,\ x,\ z+\frac{2}{3}\right]$ | `[2]` |
| 3 | $\left[-y,\ x-y,\ z+\frac{1}{3}\right]$ | `[3]` |
| 4 | $[-x,\ -y,\ z]$ | `[4]` |
| 5 | $\left[-x+y,\ -x,\ z+\frac{2}{3}\right]$ | `[5]` |
| 6 | $\left[y,\ -x+y,\ z+\frac{1}{3}\right]$ | `[6]` |